



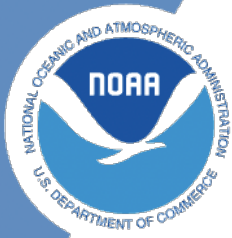
NOAA Technical Memorandum NMFS-XXX-##

Marine Ecosystems of the United States

Marine Ecosystem Division / Ecosystem
Assessment Branch

2026-09-30

U.S. DEPARTMENT OF COMMERCE
National Oceanic and Atmospheric
Administration
National Marine Fisheries Service
Northwest Fisheries Science Center



**NOAA
FISHERIES**

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1. NOAA Fisheries, Office of Science and Technology

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1 Introduction

2 About this report

The Marine Ecosystems of the United States (MEUS) report synthesizes environmental, oceanographic, ecological, and fisheries information to describe changing conditions across U.S. marine ecosystems. The report brings together a suite of indicators that characterize physical and biological conditions, ecosystem processes, and factors that may affect living marine resources and fisheries.

MEUS provides a national framework for examining patterns of ecosystem change and placing regional conditions within a broader U.S. context. Indicators are selected based on their relevance to understanding ecosystem conditions, their ability to capture meaningful changes over time, and the availability of consistent data. The report draws on established monitoring programs, research products, and other sources of information to provide a repeatable basis for tracking ecosystem conditions from year to year.

Information in this report is organized around indicators and synthesis products that describe major patterns and changes observed across U.S. marine ecosystems. Individual indicators provide information about specific environmental or ecological conditions, while synthesis products bring multiple indicators together to examine broader patterns of ecosystem change.

The report is intended to support ecosystem assessment, scientific synthesis, and fisheries management by providing accessible information on the environmental context in which U.S. fisheries operate. The indicators and synthesis presented here can also help identify emerging patterns and areas where additional investigation may be warranted.

This report was created in Quarto (<https://github.com/quarto-dev/quarto-cli/>) using the NOAA Quarto book template (<https://github.com/nmfs-opensci/NOAA-quarto-book>). The associated GitHub repository contains the data, code, figures, and other materials used to develop the report.

3 The Changing Baseline

In this section, we report indicators that describe long-term changes in the physical and ecological conditions of U.S. marine ecosystems. These indicators provide context for understanding how conditions in a given year compare with longer-term patterns and trends.

4 The Year in Marine Ecosystems

In this section, we examine the major changes and unusual conditions observed across U.S. marine ecosystems during the reporting year. Indicators are brought together to identify where conditions differed from expected patterns, how unusual those changes were, and the environmental and ecological processes that may have contributed to them. Major findings are developed into ecosystem stories that connect observations across physical and biological systems and highlight their relevance to living marine resources.

5 Implications for Living Marine Resources

In this section, we examine how major changes in marine ecosystem conditions may affect living marine resources, including changes in habitat, productivity, prey availability, species distributions, and other ecological processes. Where evidence supports a connection between ecosystem conditions and living marine resources, these relationships are described in the context of observed and emerging changes.

6 Methods and Data

This section describes the data sources, methods, and analytical approaches used to develop the indicators and synthesis presented in the report. It includes information on data processing, spatial and temporal frameworks, reference periods, indicator calculations, and considerations for interpreting the results.

7 Acknowledgments

7.1 Contributions

This report would like to acknowledge the efforts and contributions of:

7.2 Resources

This repo and GitHub Action associated with generating this report was based on the Openscapes tutorial `quarto-website-tutorial` by Julia Lowndes and Stefanie Butland.

8 Contributors

8.1 Editors

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8.2 Contributors

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References